Tide retrospective:
Sextic 1D moments

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2 May 2026

Abstract

A single Tide-loop excursion in the `laplace` seabed, generalising `Laplace/OneD/Quartic.lean` from $L(x) = x^4/24$ to $L(x) = x^6/720$. The headline theorem `Laplace.OneD.sextic_cov_affine` gives the affine covariance of the 1D pure-sextic Gibbs measure as $\mathrm{Cov}_t[ax + c, bx + d] = ab \cdot (720/t)^{1/3} \cdot \Gamma(1/2)/\Gamma(1/6)$, a rate-$t^{-1/3}$ analogue of the quartic's $t^{-1/2}$. The new file `Laplace/OneD/Sextic.lean` (396 lines) parallels `Quartic.lean` section by section. Zero `sorry`, zero `axiom`, zero `native_decide`; clean Lean diagnostics on first pass. The interesting features of this run were two tactical refinements suggested by GPT-5.5 Pro at the deliberation step — a *B+ refinement* stating the half-line moment lemma for arbitrary natural power m rather than just $2n$, and a piecewise (rather than discriminant-based) proof of the polynomial bound $x^2 \le 1 + x^6$ used for Gaussian-comparison integrability — both of which made the file genuinely cleaner than the quartic template it descended from. Like the concurrent `tide/2d-pure-quartic` excursion, this run hit a working-tree race when another agent rewrote `Laplace.lean` in the staging window; the fix was atomic re-staging plus a branch-ref re-pointing dance.

1 Setting

This Tide step is the sixth excursion outward from the same seabed: the `laplace` repo, which originally formalised the 1D anharmonic asymptotic $\mathrm{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$ (Susceptibility Primer, eq. 4.10) and has since been extended through five prior tide steps (`tide/quartic-moments`, `tide/2d-semi-degenerate`, `tide/universal-scaling`, `tide/f1-large-t`, and the concurrent `tide/2d-pure-qua`
The first four merged into `main` on 1 May (`44a6001`); this step shares its base with `2d-pure-quartic` and was branched off the latter's tip per the Tide skill's branch-hygiene rule.

The candidate was picked from the cross-tide candidate survey (`projects/automation/log/2026-05-02-tide-`
of the five candidates listed there, *Sextic 1D moments* scored as the third choice, behind the three-point κ_3 candidate (different seabed) and the 2D pure-quartic case (the parallel run). It is the cleanest test of whether the substitution-and-Γ template established by `Quartic.lean` generalises beyond a single exponent class. The pattern $L = x^{2k}/(2k)!$ predicts a rate $t^{-1/(2k-1)}$, with $k = 2$ giving the quartic's $t^{-1/2}$ and $k = 3$ giving the sextic's $t^{-1/3}$; this tide is the second data point.

The Step 2 deliberation (Claude $\leftrightarrow$ GPT-5.5 Pro) drafted three candidates: A, the full sextic package paralleling `Quartic.lean`; B, just the atomic half-line moment as a kernel; C, the general $p = 2k$ family covering quartic and sextic uniformly. Both parties backed A, with GPT making two tactical refinements that were carried into Step 3. The full deliberation is at `projects/primer/tide-log/2026-05-`

2 The thing formalised

For the centred sextic 1D potential

$$L(x) = \frac{x^6}{720}$$

and affine observables $\varphi(x) = ax + c$, $\psi(x) = bx + d$, against the 1D Gibbs measure $e^{-tL(x)}\,dx/Z(t)$:

For all $t > 0$,

$$\mathrm{Cov}_t[\varphi, \psi] \;=\; ab \cdot \left(\frac{720}{t}\right)^{1/3} \cdot \frac{\Gamma(1/2)}{\Gamma(1/6)} \;=\; ab \cdot \frac{\sqrt{\pi}\,(720/t)^{1/3}}{\Gamma(1/6)}.$$

The Lean signature:

```
theorem sextic_cov_affine {t : ℝ} (ht : 0 < t) (a b c d : ℝ) :
    gibbsCov sexticPotential t (fun x => a * x + c) (fun x => b * x + d) =
      a * b * (720 / t) ^ ((1 : ℝ) / 3) * Real.Gamma ((1 : ℝ) / 2) /
        Real.Gamma ((1 : ℝ) / 6)
```

located at `Laplace/OneD/Sextic.lean`, in namespace `Laplace.OneD`. The file commits twelve further lemmas in addition to the headline: the potential definition, two flavours of integrability, the atomic half-line moment for arbitrary natural power, even and odd full-line moments, the partition function and its positivity, even and odd expected values, and the $n = 1$ specialisation $\langle x^2 \rangle_t = (720/t)^{1/3}\Gamma(1/2)/\Gamma(1/6)$ which is the load-bearing input to `sextic_cov_affine`.

The shape $L(x) = x^{2k}/(2k)!$ means that the unique critical point $x = 0$ has $L^{(j)}(0) = 0$ for $j = 0, \ldots, 2k-1$ and $L^{(2k)}(0) = 1$: this is the rank-zero degenerate case, analogous to the quartic ($k = 2$) but more degenerate. The Hessian-based Laplace machinery from the rest of the seabed does not apply; instead the substitution $u = (t/720)^{1/6}x$ reduces every moment integral exactly to a Γ-function value, and the rate $t^{-1/3}$ falls out of the $1/(2k-1)$ formula at $k = 3$.

Numerical sanity check before formalising: `scipy.integrate.quad` agreed with the closed forms to $\sim 10^{-12}$ relative error across $t \in \{0.1, 1.0, 10.0, 100.0\}$ and $n \in \{0, 1, 2, 3\}$, for the half-line moments, the full-line even moments, the partition function, the expected values, and the affine covariance. The bonus integration-by-parts identity $\langle x^6 \rangle_t = 120/t$ (independent of all the special-function constants) provides a Gibbs-side cross-check that is also clean.

3 Proof strategy

The atomic kernel of the file is the half-line moment `integral_pow_mul_exp_neg_sextic_Ioi`, stated for arbitrary natural power m:

$$\int_0^\infty x^m e^{-tx^6/720}\,dx \;=\; \tfrac{1}{6}\left(\tfrac{720}{t}\right)^{(m+1)/6}\Gamma\left(\tfrac{m+1}{6}\right).$$

This is a direct application of Mathlib's master gamma-integral lemma `integral_rpow_mul_exp_neg_mul_rpow` with $p = 6$, $q = m$, $b = t/720$, plus a conversion between $\mathbb{N}$- and $\mathbb{R}$-powers via `rpow_natCast` and a single $(t/720)^{-(m+1)/6} \to (720/t)^{(m+1)/6}$ inversion via `inv_rpow`. Stating this for all m (the *B+ refinement*) costs nothing extra in the proof and makes the even and odd full-line cases share a single substitution argument.

From the half-line lemma the rest of the file is mechanical. The full-line even moment doubles via `integral_comp_abs` after rewriting the integrand in $|x|$-form (this uses $x^{2n} = |x|^{2n}$ and $x^6 = |x|^6$, both via `sq_abs`); the odd moment vanishes via the same parity argument the quartic file used, `Odd.neg_pow` $\langle n, \texttt{rfl}\rangle$ together with $(-x)^6 = x^6$ to flip the sign of the integrand. The partition function is the $n = 0$ specialisation of the even moment, the expected value is the ratio of the moment by the partition function (with the cancellation $(720/t)^{(2n+1)/6-1/6} = (720/t)^{n/3}$), and the affine covariance expands $(ax + c)(bx + d)$, uses linearity of `gibbsExpectation` (which itself is unfolded against `integral_add` and `integral_const_mul` after building integrability witnesses for $1, x, x^2$ against the Gibbs weight), and collapses via parity of the $\langle x\rangle$ terms to $ab\langle x^2\rangle_t$.

The integrability of $x^n e^{-tx^6/720}$ against Lebesgue measure is the only place the proof has any choices. The quartic file used an exact completing-the-square identity $t \cdot x^4/24 - x^2 = (tx^2 - 12)^2/(24t) - 6/t$ to dominate by a Gaussian with $b = 1$ and prefactor $e^{6/t}$. The sextic analogue would need to minimise a cubic $t \cdot x^6/720 - x^2$, which is messy in Lean. The route taken here, on GPT-5.5 Pro's tactical advice, is the slick polynomial bound

$$x^2 \ \leq \ 1 + x^6$$

proved piecewise: on $|x| \leq 1$, $x^2 \leq 1 \leq 1+x^6$; on $|x| \geq 1$, $x^4 \geq 1$ so $x^2 \leq x^2 \cdot x^4 = x^6 \leq 1+x^6$. Multiplying through by $t/720 > 0$ gives $t \cdot x^6/720 \geq (t/720)x^2 - t/720$, hence $e^{-tx^6/720} \leq e^{t/720} \cdot e^{-(t/720)x^2}$, dominated by a Gaussian of width $t/720$. Mathlib's `integrable_rpow_mul_exp_neg_mul_sq` handles the dominator directly with $b = t/720$; the rest is bookkeeping. The piecewise lemma is nine lines; its quartic counterpart was thirteen.

4 Roadblocks and resolutions

None of mathematical substance. The proof typechecked clean on the first compile after a one-line tactic fix (`rcases le_or_lt` → `rcases le_total`; `le_or_lt` no longer exists in the version of Mathlib pinned by the seabed) and a one-line cleanup of an unused `push_cast` that had been a vestige of copy-pasting from the quartic file. The candidate had been picked precisely so that the seabed's machinery would already cover the moves; the run paid the cost of that prediction.

Concurrent-agent staging race. Same root cause as the parallel 2D pure-quartic Tide. Multiple times during this run, the working-tree copy of `Laplace.lean` was overwritten by the other agent who was also adding an import for their new file. The interesting variant, in this run, was that the `git commit` fired while the working tree was on the *wrong branch*: at the moment of commit, the parallel agent had checked out their `tide/2d-pure-quartic` branch in the shared working tree, so my Sextic commit landed on *their* branch rather than my own. The fix was

```
git branch -f tide/sextic-moments HEAD       # claim the commit
git checkout tide/sextic-moments
git branch -f tide/2d-pure-quartic HEAD~1    # restore their tip
```

which by happy accident produced exactly the linear-stack arrangement the Tide skill's branch-hygiene rule requires (Sextic on top of PureQuartic on top of `main`), but it was a recovery move rather than a deliberate one. After the parallel run later added a trailing `Laplace.lean` import-list fix-up commit and a retrospective commit on top of their original tip, this branch was rebased onto the new tip via `git rebase tide/2d-pure-quartic`, which applied without conflict.

The structural lesson is the same as the parallel retrospective concluded: concurrent Tide agents need filesystem isolation. `git worktree` per Tide branch is the right answer.

Tactical advice from GPT-5.5 Pro that paid off. Two pieces of GPT-5.5 Pro's deliberation-stage response were carried into the proof and each paid for itself.

- *The B+ refinement.* GPT noted that on Ioi 0, parity is irrelevant, so stating the half-line moment for arbitrary natural power m costs nothing extra and makes the file's even/odd full-line story share a single substitution argument. This is the kind of strict-generalisation-for-free that Tide-loop deliberation is for; carrying it in saved a near-duplicate half-line argument for the odd case.
- *Piecewise integrability.* GPT pre-empted the obvious sin (which I had drafted) of proving $x^2 \leq 1 + x^6$ via a cubic-discriminant argument:

  > **GPT-5.5 Pro:** Your proposed bound $x^2 \leq 1 + x^6$ is excellent, but in Lean I would **not** prove it via discriminant/minimum. I would prove it piecewise: if $|x| \leq 1$, then $x^2 \leq 1 \leq 1 + x^6$; if $1 \leq |x|$, then $x^2 \leq x^6$, hence $x^2 \leq 1 + x^6$. That will be much friendlier than a cubic-minimum argument.

  The piecewise version that landed is nine lines, three of which are `nlinarith`-with-hint closers; the discriminant version would have been a forty-line minimisation by hand. This was the highest-value tactical advice in the deliberation.

Beyond these two, no proof-side roadblocks: the file is essentially a substitute-and-rename of `Quartic.lean` with the integrability lemma swapped out and the half-line lemma generalised in m.

5 What was Mathlib, what was new

Mathlib provided. `integral_rpow_mul_exp_neg_mul_rpow` for the master half-line gamma integral. `integrable_rpow_mul_exp_neg_mul_sq` for the Gaussian dominator. `integral_comp_abs` for the even-symmetric reduction to the half line. `integral_neg_eq_self` for the odd parity argument. The standard `integral_add` / `integral_const_mul` / `Integrable.const_mul/add` lemmas for the affine-observable expansion. `rpow_natCast`, `inv_rpow`, `rpow_neg`, `Real.rpow_add` for exponent algebra. `Real.Gamma_pos_of_pos` for partition-function positivity.

The seabed provided. `Laplace.Gibbs` (`partitionFunction`, `gibbsExpectation`, `gibbsCov` for the 1D Gibbs interface), and `Laplace/OneD/Quartic.lean` as the structural template. No new seabed-side infrastructure was needed.

What was new in this tide. The 1D pure-sextic potential `sexticPotential`, the Gaussian-comparison integrability `sextic_integrable_pow` (with the piecewise polynomial bound `sq_le_one_add_pow_six` as a private helper), the half-line moment `integral_pow_mul_exp_neg_sextic_Ioi` (in the B+ form for arbitrary natural power), the full-line moments (`sextic_moment_even`, `sextic_moment_odd`), the partition function (`sextic_partition`, `sextic_partition_pos`), the expected values (`sextic_expected_value_eve` and the headline `sextic_cov_affine`. There is no novel mathematical content; what is new is the substitution-into-the-template that confirms the substitution template ports cleanly to a second exponent class. As the `tide/2d-pure-quartic` retrospective predicted in its *Lessons*: "the next test is whether [the factorisation template] ports across the sextic side; if so, the abstraction is genuinely the right shape." It does, and it is.

6 Lessons

- *Specific tactical advice from GPT-5.5 Pro at the deliberation stage transfers cleanly into Step 3.* The B+ refinement and the piecewise integrability bound were both proposed at Step 2 and carried verbatim into the final proof. This is the right division of labour: the deliberation stage is where "how you should phrase the lemma" and "how you should prove the polynomial inequality" get decided, before any Lean is written. The cost was a single 5-minute consult; the value, fifty lines of avoided proof.

- *The substitution-and-*Γ *template is robust at* $k = 3$. The $L = x^{2k}/(2k)!$ family produces, at each k, a rate $t^{-1/(2k-1)}$ covariance with constants in $\Gamma((2n+1)/(2k))/\Gamma(1/(2k))$ and a proof that mirrors the previous k section by section. With $k = 2$ and $k = 3$ both done in essentially the same shape, candidate C from the deliberation (the unifying `p = 2k` parameterisation) becomes substantially more attractive than it did at $k = 2$ alone — the case for "just refactor and parameterise" has gone from one data point to two.

- *Concurrent Tide agents in the same checkout fight over `Laplace.lean`.* Same lesson as the parallel retrospective draws. Filesystem isolation via `git worktree` is the right structural fix; ad-hoc atomic re-staging is at best a recovery move.

- *Piecewise polynomial bounds beat discriminant-based ones in Lean.* The `nlinarith`-with-hint closers do not like cubic minima, but a case split on $|x| \le 1$ vs $|x| \ge 1$ followed by a single `nlinarith` per branch is robust. Worth lifting to the seabed's `CLAUDE.md` as a tactic note.

7 Follow-ups

- *The `p = 2k` parameterisation, now plausible.* With `Quartic.lean` ($k = 2$) and `Sextic.lean` ($k = 3$) now both in place and structurally near-identical, the unifying lemma

$$\int_0^\infty x^m e^{-tx^{2k}/(2k)!}\,dx \;=\; \tfrac{1}{2k}\left(\tfrac{(2k)!}{t}\right)^{(m+1)/(2k)}\Gamma\left(\tfrac{m+1}{2k}\right)$$

  is now reachable in roughly the same proof body as the present file, just with k generic. Best opened as a future Tide step: not a refactor of the existing files (they stay as worked examples), but a new `Laplace/OneD/PolyEven.lean` (or similar) that proves the parameterised identity once and recovers the quartic and sextic specialisations as corollaries. Candidate B and C from the present deliberation would land together as part of that step.

- *Quartic–sextic mixed potential.* $L(x, y) = x^4/24 + y^6/720$, with rates $t^{-1/2}$ in x and $t^{-1/3}$ in y. Mechanical given the present file plus `Laplace/TwoD/PureQuartic.lean` from the parallel Tide, particularly if the additively-separable factorisation refactor (deferred in that retrospective) lands first.

- *Mathlib upstreaming for polynomial-times-Gibbs integrability.* `quartic_integrable_pow` and `sextic_integrable_pow` are the same shape: $\int x^n \exp(-b \cdot x^{2k})$ for $b > 0, k \ge 2, n \in \mathbb{N}$. With two concrete uses, the case for promoting a generic version to `lean-common` (or upstream into Mathlib alongside the existing `integrable_rpow_mul_exp_neg_mul_sq`) is now stronger than at one use.

- *`\leanref` tagging.* If the primer gains a sextic-rate paragraph in its degenerate-case discussion, tag it with `\leanref{sextic_cov_affine}`. Currently no such paragraph exists; the sextic case is implicit in the $L = x^{2k}/(2k)!$ generalisation that the primer alludes to but does not spell out. Worth a one-paragraph appendix addition the next time the primer's degenerate-case section is touched.

- *Worktree-isolated Tide runs.* Same as the `tide/2d-pure-quartic` retrospective notes: a small wrapper that creates a `git worktree` per Tide branch and runs the agent against that worktree would prevent the staging race. Worth adding to the Tide skill as a preflight setup step.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the B+ refinement of the half-line moment lemma and the piecewise proof of the polynomial bound $x^2 \leq 1 + x^6$. Both were carried into the final proof, and both were the right call. The full consult is preserved at `projects/primer/tide-log/gpt55_tide_sextic_v1.md`. The tide log itself, with candidate drafts, vote, numerical check, and result, is at `projects/primer/tide-log/2026-05-02-tide-sextic-moments.md`. The parallel `tide/2d-pure-quartic` run shared the seabed and the integration race; its retrospective is the immediate predecessor to this one and recorded the same lessons from the other side.